Baseball Analytics Using Machine Learning

ECE1513

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Introduction to Machine Learning – Winter 2026

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Problem & Dataset

Goal: Predict team wins in season t from previous-season stats — **supervised regression**.

Dataset

- **Lahman Baseball Database** — `Teams`
- $\sim$30 teams/yr $\times$ $\sim$20 seasons $\approx$ **600 rows**
- Years 2000–2025, **excl. 2020** (COVID)

Temporal split (no data leakage)
Train: 2001–2019 ($\sim$570 rows)
Test: 2021–2025 ($\sim$150 rows)

Features lagged one season per franchise.

Pipeline (increasing complexity)

1. Pythagorean expectation $\left(\text{Win\%} = \frac{R^2}{R^2+\text{RA}^2}\right)$ – *no learning*
2. Ridge Regression – *linear*
3. SVR – *kernel*
4. Neural Network – *deep*

15 Features (previous-season stats)

Batting (7)

R	Runs scored	BB	Walks
H	Hits	SO	Strikeouts
HR	Home runs	SB	Stolen bases
		HBP	Hit by pitch

Pitching (6)

RA	Runs allowed	BBA	Walks allowed
ERA	Earned run avg	SOA	Strikeouts (P)
HA	Hits allowed	HRA	HR allowed

Fielding (2)

E	Errors	FP	Fielding pct

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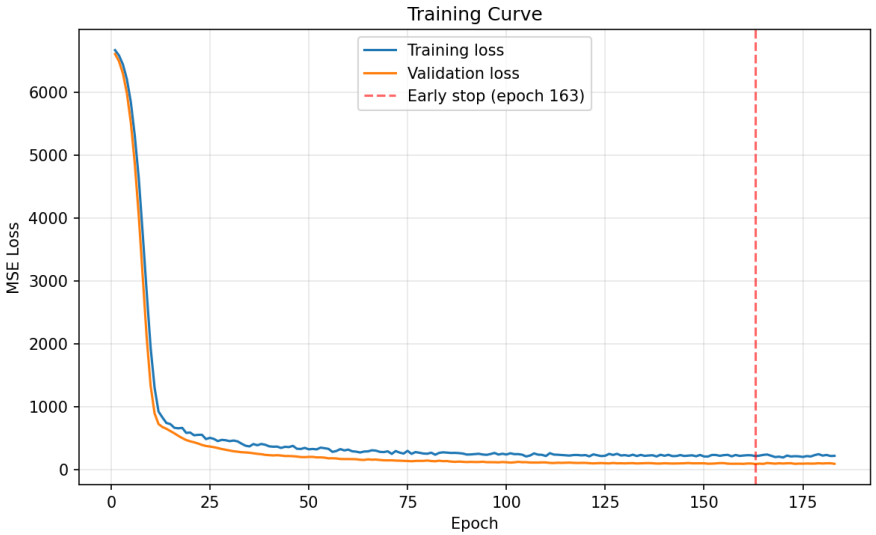
Neural Network Results

FNN · $15 \to 64 \to 32 \to 1$ · 3,137 params · ReLU + linear output · MSE loss · Adam

Training curve



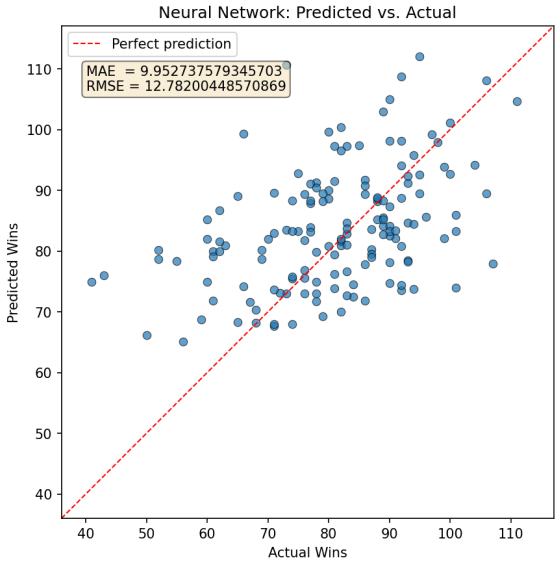
Early stopped at epoch 249 / 269 — no overfitting gap

- Mini-batch SGD ($B = 32$) + Adam
- Weight decay $\lambda = 10^{-4}$, Dropout $p = 0.2$
- $\sim$570 train samples, 15% for validation

vs. Pythagorean baseline (no learning)

MAE	RMSE
9.85	**12.59**
vs 10.73	vs 13.48

Predicted vs. actual (test set)



$\sim$8% MAE improvement

over no-learning baseline

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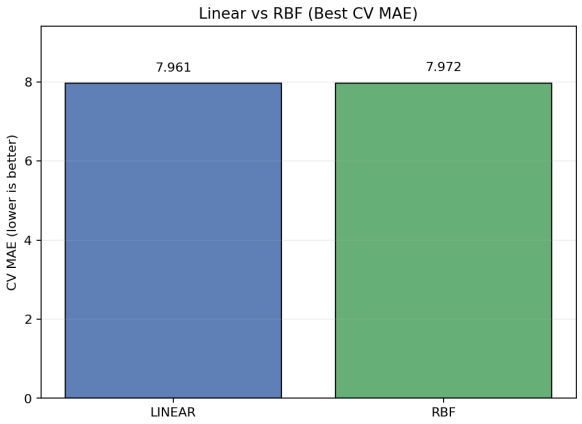
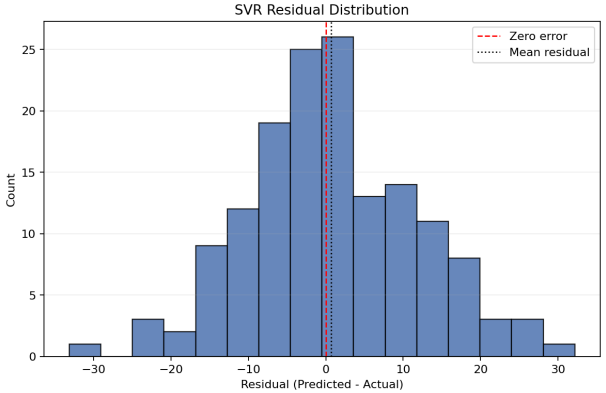
SVR Results

SVR · kernels: linear / RBF · GridSearchCV (MAE) · best: linear, $C = 0.1$, $\epsilon = 0.1$

Kernel comparison (CV MAE)



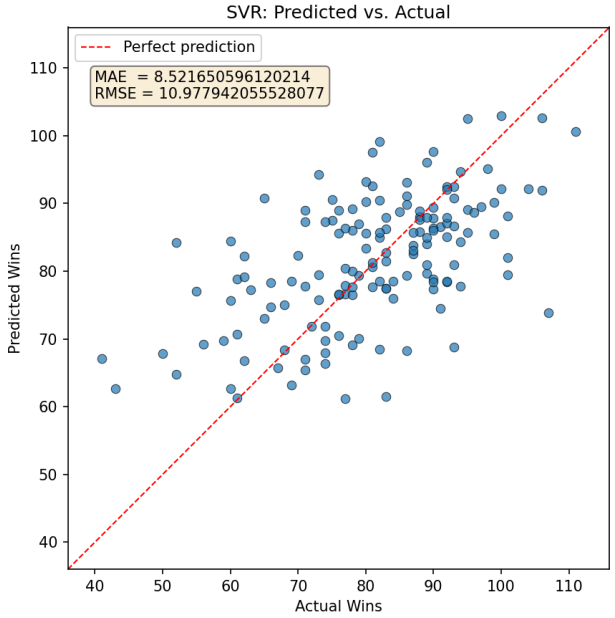
Residual distribution (test set)



vs. Pythagorean baseline (no learning)

MAE	RMSE
8.52	**10.98**
vs 10.73	vs 13.48

Predicted vs. actual (test set)



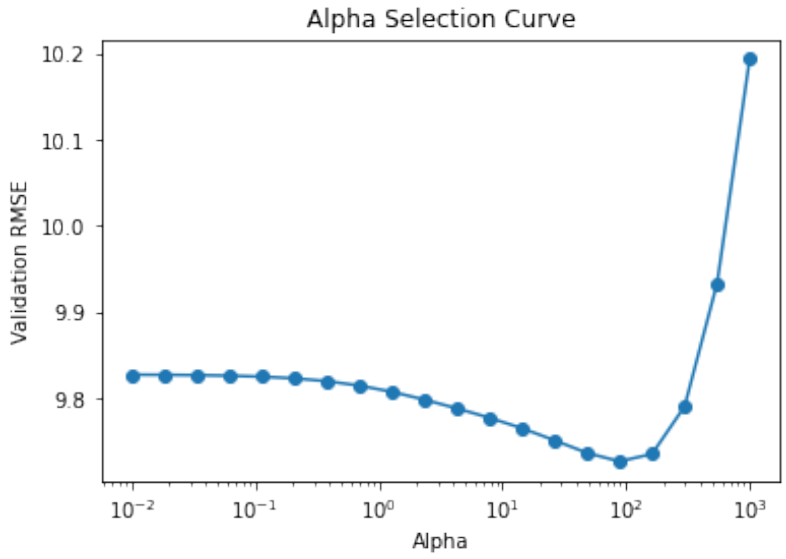
~21% MAE improvement

over no-learning baseline

Regression Results

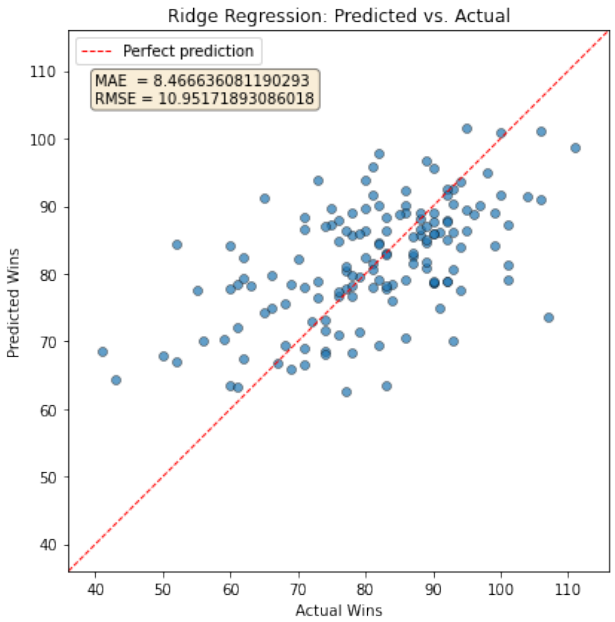
Regression · Best alpha: 88.6

Alpha Selection Curve



vs. Pythagorean baseline (no learning)	
MAE	RMSE
8.47	**10.95**
vs 10.73	vs 13.48

Predicted vs. actual (test set)



~21% MAE improvement

over no-learning baseline

Comparison & Conclusions

Model	MAE	RMSE	Params	MAE Δ
Pythagorean	10.73	13.48	0 (formula)	—
Ridge ($\alpha = 88.6$)	**8.47**	**10.95**	15	-21%
SVR (linear, $C = 0.1$)	8.52	10.98	#SVs	-21%
Neural Net	9.85	12.59	3,137	-8%

Why do simpler models win?

- **Limited data** ($\sim$570 train rows): more parameters $\Rightarrow$ higher memorization risk
- **Bias–variance trade-off:** Ridge adds small bias, greatly reduces variance
- **SVR best kernel = linear** $\Rightarrow$ the runs$\rightarrow$wins relationship is approximately linear; RBF adds capacity the data doesn't need
- **NN:** universal approximation guarantees capacity , but SGD on small data may not find the optimum

Key Takeaways

1. All ML models beat the no-learning baseline
2. ℓ_2 **regularization** is the single most effective technique on small data
3. Model complexity must match data size — more params $\neq$ better
4. Season-to-season randomness (injuries, trades) sets a floor $\approx$ Bayes error

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